

HYDROGENATED NBR: PATENT LANDSCAPE ON LOW-MOONEY HNBR, METATHESIS-BASED MOLECULAR WEIGHT REDUCTION, AND FUNCTIONALIZED HNBR APPLICATIONS

1. ABSTRACT

This report synthesizes a set of patents centered on hydrogenated nitrile rubber (HNBR) synthesis, molecular-weight control, and downstream formulation use. The target material is defined as nitrile-butadiene rubber (NBR) that has been partially or fully hydrogenated to reduce residual unsaturation while retaining the nitrile-containing backbone needed for oil, heat, and ozone resistance. Across the selected patents, two major synthetic strategies dominate: (1) post-polymerization hydrogenation of NBR using homogeneous catalysts, especially rhodium- and ruthenium-based systems, and (2) metathesis-assisted molecular-weight reduction of NBR prior to or concurrent with hydrogenation to produce lower-Mooney, narrower-distribution HNBR. Several patents emphasize that conventional HNBR suffers from high Mooney viscosity and broad molecular-weight distribution, motivating catalyst systems and process conditions that lower Mw, PDI, and residual double-bond content without introducing unwanted functional groups or excessive gelation.

The landscape includes patents focused on low molecular weight HNBR obtained by metathesis degradation followed by hydrogenation, as well as a single-step simultaneous metathesis/hydrogenation route using ruthenium or osmium complexes. Other patents address partially hydrogenated nitrile rubbers with a defined ratio of solution viscosity to molecular weight, and microgel-containing vulcanizable compositions based on HNBR for improved high-temperature mechanical performance. A later patent extends HNBR into battery technology by using epoxy-functionalized, optionally hydrogenated nitrile rubber as a fluorine-free cathode binder. Across the portfolio, the major process trends are use of homogeneous metal catalysts, optional co-olefins, solvent-based cement processing, controlled hydrogenation levels, and formulation additives such as carbon black, peroxide, unsaturated carboxylic acids, and microgels. The evidence collectively shows that HNBR synthesis is being optimized not only for saturation level, but also for processability, molecular-weight narrowing, and compatibility with demanding end-use applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8124234B2
- Patent Title: Polyurethane compositions and articles prepared therefrom, and methods for ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not directed to HNBR synthesis. It discloses polyurethane-based compositions containing an elastomer rubber component, a thermoplastic polyurethane, and a polydiene- or polydiol-based polyurethane for adhesion and surface-property improvement in automotive and related articles. - 55 to 80 weight percent olefin-based polymer — Claim 1 - 15 to 35 weight percent thermoplastic polyurethane — Claim 1 - 5 to 10 weight percent polydiene-based or polydiol-based polyurethane — Claim 1 - 160°C — Comparative Example 1 - 2 minutes — Comparative Example 1 - 5 minutes — Comparative Example 1 - 170°C — Comparative Example 4 - 0.10 wt percent Luperox 101 peroxide — Comparative Example 4 - 5 minutes — Comparative Example 4 - 180°C — Comparative Example 6 - 80/20 TPU/ENGAGE mass ratio — Comparative Example 10 - 140°C — Comparative Example 12 - 6895 kPa — Comparative Example 12 - 275790 kPa — Comparative Example 12 - 3 minutes — Comparative Example 12 - 7 minutes — Comparative Example 12 - 120°C — Comparative Example 13 - 500 hours — Comparative Example 13 - 120°C — Comparative Example 14 - 1.02 mm sheet thickness — Comparative Example 14 - 140°C, 170°C, 170°C, 170°C — Comparative Example 15 - 175°C, 185°C, 190°C — Comparative Example 15 - 250°C — Comparative Example 16 - 18°C — Comparative Example 16 - 110°C — Comparative Example 17 - 15 minutes — Comparative Example 17 - 10 mg Grubbs II catalyst — not HNBR-related example context - 1 mg LiBr — Example 7 - 1 mg CsBr — Example 8

Relevant Experimental Evidence - Example a: The text discusses automotive sheet/skin requirements such as low gloss, heat resistance, and adhesion, but does not disclose HNBR synthesis. - Comparative Example 4: Peroxide-assisted functionalization of an ethylene-octene elastomer was demonstrated, not HNBR chemistry. - Comparative Example 10: Functionalized polyolefin/TPU blends showed improved dispersion and smaller particle size with amine functionality. - Comparative Example 13: Compatibilized sheet compositions showed surface energy, gloss, and heat-aging performance trends. - Comparative Example 16: Extruder-based preparation of amine- and hydroxyl-functionalized polyolefin compatibilizers was shown.

Technical Relevance This patent is only indirectly relevant to Hydrogenated NBR because it demonstrates polymer blend and adhesion concepts, not HNBR synthesis. It provides no direct route to hydrogenated nitrile rubber, but it is useful as a formulation reference for elastomer/TPU compatibility.

Patent 2

- Patent Number: EP2125918B1
- Patent Title: Functionalized olefin polymers, compositions and articles prepared therefrom, ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of functionalized polyolefin polymers by grafting an olefin-based polymer with an amine-reactive compound, then blending the grafted polymer in solid form with a primary-secondary diamine to form an imide-functionalized polyolefin. The examples show melt functionalization and subsequent property evaluation in TPU and polycarbonate blends. - 160°C — Comparative Example 1 - 2 minutes — Comparative Example 1 - 1.61 g N-ethyl-ethylenediamine — Comparative Example 1 - 18.2 mmol N-ethyl-ethylenediamine — Comparative Example 1 - 5 minutes — Comparative Example 1 - 170°C — Comparative Example 4 - 1.0 g maleic anhydride — Comparative Example 4 - 0.10 wt percent Luperox 101 peroxide — Comparative Example 4 - 5 minutes — Comparative Example 4 - 1.21 g ethanolamine — Comparative Example 4 - 19.8 mmol ethanolamine — Comparative Example 4 - 5 more minutes — Comparative Example 4 - 170°C — Comparative Example 8 - 100 rpm — Comparative Example 8 - 45 g ENGAGE 8130 — Comparative Example 8 - 1.0 g maleic anhydride — Comparative Example 8 - 10.2 mmol maleic anhydride — Comparative Example 8 - 0.0504 g active Luperox 101 peroxide — Comparative Example 8 - 0.7 mmol RO· — Comparative Example 8 - 5 minutes — Comparative Example 8 - 1.2 g ethanolamine — Comparative Example 8 - 19.7 mmol ethanolamine — Comparative Example 8 - 2 minutes — Comparative Example 8 - 300 MHz — Comparative Example 8 - 80°C — Comparative Example 8 - 170°C — Comparative Example 9 - 1.5 g maleamic acid — Comparative Example 9 - 9.4 mmol maleamic acid — Comparative Example 9 - 0.0504 g active Luperox 101 peroxide — Comparative Example 9 - 0.7 mmol RO· — Comparative Example 9 - 5 minutes — Comparative Example 9 - 300 MHz — Comparative Example 9 - 80°C — Comparative Example 9 - 80/20 TPU/ethylene-octene ratio — Comparative Example 10 - 180°C — Comparative Example 10 - 5 minutes — Comparative Example 10 - 0.84 ± 0.79 μm mean particle size — Comparative Example 10 - 0.35 ± 0.28 μm mean particle size — Comparative Example 10 - 0.42 ± 0.32 μm mean particle size — Comparative Example 10 - 0.11 ± 0.10 μm mean particle size — Comparative Example 10 - 50 wt percent amine-functionalized elastomer + 50 wt percent ENGAGE 8100 — Comparative Example 11 - 25 wt percent amine-functionalized elastomer + 75 wt percent

ENGAGE 8100 — Comparative Example 11 - 12 wt percent amine-functionalized elastomer + 88 wt percent ENGAGE 8100 — Comparative Example 11 - 180°C — Comparative Example 11 - 5 minutes — Comparative Example 11 - 140°C — Comparative Example 12 - 6895 kPa — Comparative Example 12 - 275790 kPa — Comparative Example 12 - 3 minutes, 3 minutes, 7 minutes — Comparative Example 12 - 23°C — Comparative Example 12 - 50 percent RH — Comparative Example 12 - 1.5 to 1.6 mm thickness — Comparative Example 12 - 1.02 mm sheet thickness — Comparative Example 14 - 140°C, 170°C, 170°C, 170°C — Comparative Example 15 - 175°C, 185°C, 190°C — Comparative Example 15 - 2.5 wt percent Amplify GR216 — Comparative Example 15 - 60.5 wt percent ENR 7086.01 — Comparative Example 15 - 37 wt percent Pellethane 2103-70A — Comparative Example 15 - 250°C — Comparative Example 16 - 18°C — Comparative Example 16 - 110°C — Comparative Example 17 - 15 minutes — Comparative Example 17 - 10 mg Grubbs II catalyst — Example 7 - 1 mg LiBr — Example 7 - 1 mg CsBr — Example 8

Relevant Experimental Evidence - Comparative Example 1: Maleic-anhydride-grafted polyolefin was converted to an imide-functionalized product using N-ethyl-ethylenediamine. - Comparative Examples 1A and 1B: Alternative diamines caused crosslinking and insoluble gel formation. - Comparative Example 4: Peroxide-assisted grafting followed by ethanolamine reaction produced hydroxy-functionalized polyolefin. - Comparative Example 10: Amine-functionalized elastomer gave the smallest TPU dispersed-particle size among the tested blends. - Comparative Example 12: Functionalized polyolefin films were prepared for peel testing against polycarbonate. - Comparative Example 18: PU glue adhesion to functionalized polyolefin/PET fabric blends reached 7.2 to 9.2 N/mm depending on composition.

Technical Relevance This patent is relevant to HNBR only as an adjacent functionalization/compatibilization reference, not as an HNBR synthesis disclosure. It demonstrates melt-functionalization chemistry and elastomer compatibility strategies that could be conceptually useful for HNBR-based blends or binders.

Patent 3

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a process for preparing low molecular weight HNBR by subjecting nitrile rubber to metathesis in the presence of a co-olefin and then hydrogenating the degraded polymer. The process is designed to reduce Mw, Mooney viscosity, and polydispersity while preserving the HNBR microstructure. - 30,000 to

250,000 — Claim 1 MW range - less than 10 — Claim 1 Mooney viscosity - less than 2.5 — Claim 1 MWD/PDI - 40,000 to 220,000 — Claim 2 MW range - less than 2.3 — Claim 3 PDI - 200 g rubber — Example 1 - 1133 g MCB — Example 1 - 15 wt.-% solid — Example 1 - 3 times — Example 1 - 6.9×10^5 Pa — Example 1 - 200 g rubber — Example 2 - 1133 g MCB — Example 2 - 15 wt.-% solid — Example 2 - 3 times — Example 2 - 6.9×10^5 Pa — Example 2 - 75 g rubber — Example 3 - 1175 g MCB — Example 3 - 6 wt.-% solid — Example 3 - 3 times — Example 3 - 6.9×10^5 Pa — Example 3 - 75 g rubber — Example 4 - 1175 g MCB — Example 4 - 150 g 1-hexene — Example 4 - 3 times — Example 4 - 80°C — Table 1 - 0.05 phr catalyst load — Table 1 - 0.10 phr catalyst load — Table 1 - 0.25 phr catalyst load — Table 1 - 27 kg/mol Mn — Table 2 - 54 kg/mol Mw — Table 2 - 3.4 PDI starting substrate — Table 2 - 2.0 PDI metathesized product — Table 2 - 1.90 minimum PDI — Table 2 - 73 Mooney viscosity — Table 2 starting substrate - 43 Mooney viscosity — Table 2 Experiment 1 - 28 Mooney viscosity — Table 2 Experiment 2 - 3 Mooney viscosity — Table 2 Experiment 3 - 31 Mooney viscosity — Table 2 Experiment 4

Relevant Experimental Evidence - Example 1: Ethylene-degassed NBR cement was used as a metathesis substrate under full agitation. - Example 2: A second cement-based metathesis run was performed under the same ethylene pressure conditions. - Example 3: Lower-solids cement was processed similarly, showing the method is applicable at reduced concentration. - Example 4: 1-hexene was used as co-olefin, demonstrating alternative chain-transfer/metathesis conditions. - Table 2: Metathesis reduced Mw and Mooney viscosity substantially, with PDI dropping to 1.90 in one experiment.

Technical Relevance This is a core HNBR synthesis patent because it directly addresses the main problem of high Mooney viscosity in HNBR by combining metathesis degradation with hydrogenation. It provides explicit evidence that lower molecular weight and narrower molecular-weight distribution can be achieved while retaining HNBR functionality.

Patent 4

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent also discloses low molecular weight HNBR prepared by metathesis degradation of nitrile rubber followed by hydrogenation. It emphasizes controlled molecular-weight reduction using co-olefin-assisted metathesis and reports the resulting reduction in Mw and Mooney viscosity. - 30,000 to 250,000 — Claim 1 MW range - less than 10 — Claim 1 Mooney viscosity - less than 2.5 — Claim 1 MWD/PDI - 40,000 to 220,000 — Claim 2 MW range - less than 2.3 — Claim 3 PDI - 200 g rubber —

Example 1 - 1133 g MCB — Example 1 - 15 wt-% solid — Example 1 - 3 times — Example 1 - 6.9×10^5 Pa — Example 1 - 200 g rubber — Example 2 - 1133 g MCB — Example 2 - 15 wt-% solid — Example 2 - 3 times — Example 2 - 6.9×10^5 Pa — Example 2 - 75 g rubber — Example 3 - 1175 g MCB — Example 3 - 6 wt-% solid — Example 3 - 3 times — Example 3 - 6.9×10^5 Pa — Example 3 - 75 g rubber — Example 4 - 1175 g MCB — Example 4 - 150 g 1-hexene — Example 4 - 3 times — Example 4 - 80°C — Table 1 - 0.05 phr catalyst load — Table 1 - 0.10 phr catalyst load — Table 1 - 0.25 phr catalyst load — Table 1 - 27 kg/mol Mn — Table 2 - 54 kg/mol Mw — Table 2 - 3.4 PDI starting substrate — Table 2 - 2.0 PDI metathesized product — Table 2 - 1.90 minimum PDI — Table 2 - 73 Mooney viscosity — Table 2 starting substrate - 43 Mooney viscosity — Table 2 Experiment 1 - 28 Mooney viscosity — Table 2 Experiment 2 - 3 Mooney viscosity — Table 2 Experiment 3 - 31 Mooney viscosity — Table 2 Experiment 4

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis was run on a 15 wt-% cement solution. - Example 2: A repeat run confirmed the same metathesis/hydrogenation workflow. - Example 3: Lower-solids cement was processed successfully. - Example 4: 1-hexene co-olefin was introduced, showing flexibility in chain-transfer reagent choice. - Table 2: The metathesized product showed lower Mw and lower Mooney viscosity than the starting substrate.

Technical Relevance This patent is directly relevant to HNBR synthesis because it targets the same low-viscosity, low-PDI product space as US7919563B2. It reinforces the importance of metathesis-based molecular-weight control before or during hydrogenation.

Patent 5

- Patent Number: EP2057205B1
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses segment-interchange metathesis between ethylenically unsaturated polymers, including partially hydrogenated nitrile/butadiene rubber and other unsaturated polymers, to form meta-block copolymers. HNBR appears here as a reagent polymer rather than the final target, but the process uses Grubbs II or tungsten-based metathesis catalysts in solution. - 0.25 g partially-hydrogenated polybutadiene — Example 1 - 0.25 g partially-hydrogenated nitrile/butadiene rubber — Example 1 - 95 °C — Example 1 - 8 mg Grubbs II metathesis catalyst — Example 1 - 1 hour — Example 1 - 0.25 g unsaturated polyurethane — Example 2 - 0.25 g poly(ethylene-co-butadiene) — Example 2 - 105 °C — Example 2 - 9 mg Grubbs II metathesis catalyst — Example 2 - 1 hour — Example 2 - 1.05 g fumaryl-modified polycarbonate — Example 3 - 0.45 g poly(ethylene-co-butadiene) — Example 3 - 105 °C — Example 3 - 9 mg Grubbs II metathesis catalyst — Example 3 - 1 hour —

Example 3 - 0.25 g fumaryl-modified polycarbonate — Example 4 - 0.25 g poly(ethylene-co-butadiene) — Example 4 - 105 °C — Example 4 - 9 mg Grubbs II metathesis catalyst — Example 4 - 1 hour — Example 4 - 0.25 g unsaturated poly(ethylene oxide) — Example 5 - 0.25 g poly(ethylene-co-butadiene) — Example 5 - 105 °C — Example 5 - 9 mg Grubbs II catalyst — Example 5 - 1 hour — Example 5 - 30 ml tetrahydrofuran — Example 5 - 0.25 g fumaryl-modified polycarbonate — Example 6 - 0.25 g poly(ethylene-co-octene-co-butadiene) — Example 6 - 105 °C — Example 6 - 9 mg Grubbs II metathesis catalyst — Example 6 - 1 hour — Example 6 - 0.10 g poly(ethylene-co-butadiene) — Example 7 - 0.40 g poly(ethylene-co-octene-co-butadiene) — Example 7 - 105°C — Example 7 - 8,000 ppm tungsten hexachloride catalyst — Example 7 - 13,000 ppm tri-n-butylmethyltin co-catalyst — Example 7 - 1 hour — Example 7 - 0.25 g unsaturated polyester — Example 8 - 0.25 g poly(ethylene-co-butadiene) — Example 8 - 105 °C — Example 8 - 9 mg Grubbs II catalyst — Example 8 - 1 hour — Example 8 - 30 ml tetrahydrofuran — Example 8 - 0.25 g SBS — Example 9 - 0.25 g poly(ethylene-co-butadiene) — Example 9 - 105 °C — Example 9 - 8,000 ppm tungsten hexachloride catalyst — Example 9 - 13,000 ppm tri-n-butylmethyltin co-catalyst — Example 9 - 1 hour — Example 9 - 0.25 g unsaturated fumaryl-modified polycarbonate — Example 10 - 0.25 g SBS — Example 10 - 0.25 g poly(ethylene-co-butadiene) — Example 10 - 105 °C — Example 10 - 9 mg Grubbs II catalyst — Example 10 - 1 hour — Example 10 - 50 ml tetrahydrofuran — Example 10

Relevant Experimental Evidence - Example 1: Partially hydrogenated nitrile/butadiene rubber participated in metathesis with partially hydrogenated polybutadiene to form a meta-block copolymer. - Example 2: Unsaturated polyurethane and poly(ethylene-co-butadiene) underwent segment interchange to give a meta-block copolymer. - Example 7: Tungsten hexachloride/tri-n-butylmethyltin catalyzed segment interchange between two ethylenic polymers. - Example 9: SBS and poly(ethylene-co-butadiene) formed an ordered microphase morphology consistent with meta-block copolymer formation. - Example 10: A terpolymer-like meta-block product was formed from fumaryl-modified polycarbonate, SBS, and poly(ethylene-co-butadiene).

Technical Relevance This patent is relevant because it explicitly includes partially hydrogenated nitrile/butadiene rubber as a metathesis reagent, showing that HNBR-like materials can participate in segment-interchange chemistry. It is not a primary HNBR synthesis route, but it broadens the metathesis chemistry context around hydrogenated nitrile polymers.

Patent 6

- Patent Number: US9321910B2
- Patent Title: Microgel-containing vulcanizable composition based on hydrogenated nitrile ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose HNBR synthesis; it discloses vulcanizable compositions based on hydrogenated nitrile rubber, unsaturated carboxylic acid or salts, peroxide, and microgels. The focus is on improving high-temperature mechanical properties and processing behavior of HNBR vulcanizates. - 100 parts by weight rubber — Claim 1 - 10 to 120 MU — Claim 1 Mooney viscosity - 95 to 100% — Claim 1 degree of hydrogenation - 1 to 100 parts by weight unsaturated carboxylic acids/salts — Claim 1 - 0.2 to 8 parts by weight peroxide — Claim 1 - up to 60 parts by weight microgel — Claim 2 - less than or equal to -20°C . — Claim 1 microgel Tg - less than or equal to -50°C . — Claim 3 microgel Tg - 15 to 100 MU — Claim 4 Mooney viscosity - 130°C . — Background and Summary - 150°C . — Summary and Detailed Description

Relevant Experimental Evidence - No numbered examples were detected by the parser. - The patent emphasizes microgels as a way to improve HNBR vulcanizate properties at 130°C and after hot-air aging at 150°C . - It states that microgels reduce tack and improve demolding in injection-molding processes. - The composition combines HNBR, unsaturated carboxylic acid or salt, peroxide, and microgel for high-temperature performance.

Technical Relevance This patent is highly relevant to HNBR application engineering rather than synthesis. It identifies formulation strategies that preserve or improve mechanical properties of hydrogenated nitrile rubber at elevated temperatures, which is important for practical deployment of synthesized HNBR.

Patent 7

- Patent Number: US20130345366A1
- Patent Title: Partially hydrogenated nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses partially hydrogenated nitrile rubbers with a specified degree of hydrogenation and a defined ratio of solution viscosity to molecular weight, together with a process for preparing them. The synthesis route is based on metathetical degradation of nitrile rubber followed by hydrogenation, using ruthenium- or osmium-based metathesis catalysts and optional co-olefin. - 50% to 97% — Claim 1 degree of hydrogenation - $4 \cdot 10^{-3}$ to $30 \cdot 10^{-3}$ Pasmol/kg — Claim 1 SV/Mw ratio - 52.5% to 96.5% — Claim 2 degree of hydrogenation - 55% to 96% — Claim 2 degree of hydrogenation - 57.7% to 95.5% — Claim 2 degree of hydrogenation - 60% to 95% — Claim 2 degree of hydrogenation - $5 \cdot 10^{-3}$ to $30 \cdot 10^{-3}$ Pasmol/kg — Claim 3 SV/Mw ratio - $6.5 \cdot 10^{-3}$ to $30 \cdot 10^{-3}$ Pasmol/kg — Claim 3 SV/Mw ratio - 50% to 97% — Claim 4 hydrogenation extent - 20,000 to 250,000 — Description Mw range - 1 to 50 — Description Mooney viscosity range - less than 3.0 — Description PDI - 24 to 70

— Description Mooney viscosity of nitrile rubbers used - 28 to 40 — Description preferred Mooney viscosity of nitrile rubbers used - less than 30 Mooney units — Description pre-degradation target - Mn <70 000 g/mol — Description pre-degradation target - less than 0.5 mol % vinylic double bonds — Example section - less than 0.25 mol % vinylic double bonds — Example section - 0.0001 to 1 mol % acyclic diene per mole of double bonds — Example section - 30 000 to 250 000 — Example section Mw range - 3 to 50 — Example section Mooney viscosity range - less than 2.5 — Example section PDI - 307 ppm Ru — Example section - 61 ppm Ru — Example section - 0.5 phr Grubbs I catalyst — Example section - 614 ppm ruthenium — Example section - Grubbs II catalyst — Example section - 1,3-bis(2,4,6-trimethylphenyl)-2-imidazolidenylidene)(tricyclohexylphosphine)ruthenium(phenylmethylene)dichloride — Example section - 50% to 97% — Example section degree of hydrogenation - 0.01 to 1% by weight catalyst — Description - 0.03 to 0.5% by weight catalyst — Description - 0.1 to 0.3% by weight catalyst — Description

Relevant Experimental Evidence - Example 2.17: Metathesis treatment of Therban LT 2157 reduced Mooney viscosity and Mooney relaxation values. - Table 6: The metathesis-treated rubber showed lower ML 1+1 and ML 1+4 values than the untreated reference. - Table 7: Vulcanizable mixtures were formulated with carbon black, antioxidants, zinc salts, magnesium oxide, zinc oxide, TAIC, and peroxide. - Table 8: Processing behavior was evaluated by ASTM D1646 Mooney measurements. - The description links lower Mooney and narrower SV/Mw ratio to improved processability without sacrificing vulcanizate properties.

Technical Relevance This is one of the most directly relevant patents for Hydrogenated NBR because it defines a partially hydrogenated nitrile rubber product with controlled hydrogenation and viscosity-to-molecular-weight ratio. It provides a practical route to HNBR with improved processing behavior while maintaining desirable mechanical performance.

Patent 8

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a single-step process for producing low molecular weight HNBR by simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen and a ruthenium- or osmium-based catalyst. The process is designed to give low residual double-bond content, reduced Mw, and narrower molecular-weight distribution compared with conventional HNBR. - 20,000 to 250,000 — Description Mw range - 1 to 50 — Description Mooney viscosity range - less than 3.0 —

Description PDI - 50% to 97% — Claim 1 degree of hydrogenation - 30,000 to 250,000 — Claim 1 MW range - 1 to 50 — Claim 1 Mooney viscosity range - less than 3.0 — Claim 1 MWD/PDI - 24 to 70 — Description Mooney viscosity of nitrile rubbers used - 28 to 40 — Description preferred Mooney viscosity of nitrile rubbers used - less than 30 Mooney units — Description pre-degradation target - Mn <70 000 g/mol — Description pre-degradation target - less than 0.5 mol % vinylic double bonds — Example section - less than 0.25 mol % vinylic double bonds — Example section - 0.0001 to 1 mol % acyclic diene per mole of double bonds — Example section - 30 000 to 250 000 — Example section Mw range - 3 to 50 — Example section Mooney viscosity range - less than 2.5 — Example section PDI - 307 ppm Ru — Example section - 61 ppm Ru — Example section - 0.5 phr Grubbs I catalyst — Example section - 614 ppm ruthenium — Example section - Grubbs II catalyst — Example section - 1,3-bis(2,4,6-trimethylphenyl)-2-imidazolidenylidene(tricyclohexylphosphine)ruthenium(phenylmethylene)dichloride — Example section - 50% to 97% — Example section degree of hydrogenation - 0.041 phr — Example 1 catalyst amount - 0.045 phr — Example 1 catalyst amount - 0.060 phr — Example 1 catalyst amount - 65.6 ppm — Example 1 metal content - 0.0 to 240 min — Example 1 reaction time profile - 100% to 0.4% RDB — Example 1 HNBR 1 profile - 100% to 3.5% RDB — Example 1 HNBR 2 profile - 100% to 0.7% RDB — Example 1 HNBR 3 profile - 113371 g/mol Mw — Example 1 - 48352 g/mol Mn — Example 1 - 2.34 PDI — Example 1 - 33.3 Mooney viscosity — Example 1 - 133758 g/mol Mw — Example 1 - 54595 g/mol Mn — Example 1 - 2.45 PDI — Example 1 - 41.3 Mooney viscosity — Example 1 - 195304 g/mol Mw — Example 1 - 64130 g/mol Mn — Example 1 - 3.05 PDI — Example 1 - 69.3 Mooney viscosity — Example 1

Relevant Experimental Evidence - Example 1: Three catalyst systems were compared for simultaneous hydrogenation/metathesis of nitrile rubber. - Table 2: Residual double bond content dropped from 100% to as low as 0.4% over 240 min for HNBR 1. - Table 3: The catalyst of the invention gave lower Mw and lower Mooney viscosity than the industrial benchmark. - The data show that catalyst choice strongly affects hydrogenation efficiency and molecular-weight reduction. - The process achieves low-RDB HNBR in a single step.

Technical Relevance This is a key HNBR synthesis patent because it directly claims simultaneous hydrogenation and metathesis to produce low-Mw HNBR in one step. It is especially important for process intensification and for reducing catalyst and processing burden relative to two-step routes.

Patent 9

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses catalyst-system enhancement for metathesis reactions, including nitrile rubber metathesis, by adding salts to a metathesis catalyst. The examples are general metathesis demonstrations, but the claims explicitly cover nitrile rubber metathesis with salt/catalyst combinations. - 0.01:1 to 10000:1 — Claim 1 salt:metathesis catalyst weight ratio - 1 mg LiBr — Example 7 - 1 mg CsBr — Example 8 - 10 mg Grubbs II catalyst — Example 7 - 0.3 ml chlorobenzene — Example 7 - 0.2 ml CDCl₃ — Example 7 - 0.15 ml diethyl diallylmalonate — Example 7 - 2 minutes — Example 7 - 30 min — Example 7 conversion point - 60 min — Example 7 conversion point - 10 mg Grubbs II catalyst — Example 8 - 1 mg CsBr — Example 8 - 0.3 ml chlorobenzene — Example 8 - 0.2 ml CDCl₃ — Example 8 - 0.15 ml diethyl diallylmalonate — Example 8 - 15 min — Example 8 conversion point - 30 min — Example 8 conversion point - 60 min — Example 8 conversion point - 90 min — Example 8 conversion point - 150 min — Example 8 conversion point

Relevant Experimental Evidence - Example 7: LiBr accelerated ring-closing metathesis of diethyl diallylmalonate relative to the salt-free control. - Example 8: CsBr also accelerated metathesis, though the effect was less pronounced than LiBr in the reported time window. - The patent claims that salt addition can improve metathesis catalyst activity for nitrile rubber degradation. - The examples demonstrate the general catalytic principle rather than HNBR-specific polymer data.

Technical Relevance This patent is relevant because it provides a catalyst-activation strategy that can be applied to metathesis-based NBR degradation, a key step in low-Mw HNBR synthesis. It does not itself make HNBR, but it informs catalyst optimization for the metathesis stage.

Patent 10

- Patent Number: EP3900086B1
- Patent Title: Electrode composition for a cathode of a cell of a lithium-ion battery, a ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not an HNBR synthesis patent; it discloses cathode electrode compositions for lithium-ion batteries using epoxy group-containing fluorine-free copolymers, including epoxy group-containing nitrile rubber copolymers that may be fully or partially hydrogenated. The HNBR-related content is formulation-oriented, focusing on binder performance rather than polymer preparation. - 25% by weight — Claim 1 vinyl acetate unit minimum - 5% by weight — Claim 1 ethylene unit minimum - 0.01 to 30% by weight — Claim 1 epoxy group-containing monomer unit - 5 to 60% by weight — Claim 1 α,β -unsaturated nitrile unit - 10 to 94.99% by weight — Claim 1 conjugated diene unit, optionally fully or partially hydrogenated - 0.01 to 30% by weight — Claim 1 epoxy group-containing monomer unit - LiPF₆ — Background electrolyte salt - LiAsF₆ — Background

electrolyte salt - LiCF₃SO₃ — Background electrolyte salt - LiCO₄ — Background electrolyte salt - acetonitrile — Background solvent - tetrahydrofuran — Background solvent - ethylene carbonate — Background solvent - ethyl methyl carbonate — Background solvent - dimethyl carbonate — Background solvent - vinylene carbonate — Background solvent - propylene carbonate — Background solvent

Relevant Experimental Evidence - No numbered examples were detected by the parser. - The claims explicitly include epoxy group-containing nitrile rubber copolymers with partially or fully hydrogenated diene units. - The patent positions hydrogenated nitrile rubber as a fluorine-free binder option for cathodes. - The description emphasizes adhesion, cohesion, flexibility, and electrochemical stability in battery electrodes.

Technical Relevance This patent is relevant to hydrogenated NBR as an application patent, not a synthesis patent. It shows that hydrogenated nitrile rubber can be functionalized and used as a binder in lithium-ion battery cathodes, expanding the utility space for HNBR-derived materials.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7919563B2, US7772328B2, US20130345366A1, and US9150669B2 all target low-Mooney, low-PDI HNBR, but US9150669B2 is the only one explicitly claiming simultaneous hydrogenation and metathesis in a single step.
- US7919563B2 and US7772328B2 emphasize metathesis degradation followed by hydrogenation, with co-olefin use and cement/solvent processing; US20130345366A1 extends this by defining a specific SV/Mw ratio window for partially hydrogenated nitrile rubbers.
- Catalyst families differ across the HNBR synthesis patents: US7919563B2 and US7772328B2 discuss Hoveyda, Arlt, and Wilkinson-type hydrogenation catalysts; US20130345366A1 and US9150669B2 focus on ruthenium/osmium metathesis catalysts, including Grubbs-type systems.
- Reaction conditions in the low-MW HNBR patents are broadly similar in using solvent-based cements and elevated temperatures around 80°C to 105°C, but the exact catalyst loadings vary from 0.041 to 0.060 phr in US9150669B2 and 0.05 to 0.25 phr in US7919563B2/US7772328B2.
- US9321910B2 and EP3900086B1 are not synthesis patents; they show how HNBR is leveraged in vulcanizates and battery binders, respectively, highlighting the downstream value of controlled hydrogenation and functionalization.
- EP2057205B1 is the only patent in the set that uses partially hydrogenated nitrile/butadiene rubber as a metathesis reagent for meta-block copolymer formation rather than as the final HNBR product.
- US8957161B2 is a catalyst-additive patent that can support HNBR synthesis indirectly by improving metathesis activity, but it does not itself disclose HNBR product properties.

4. CONCLUSION

The evidence shows that the most effective HNBR synthesis strategies in this patent set are metathesis-based molecular-weight reduction of NBR followed by hydrogenation, and in one case simultaneous metathesis/hydrogenation in a single step. The most useful product targets are low-Mw HNBR with Mw roughly 20,000 to 250,000, Mooney viscosity below about 50, and reduced PDI, with hydrogenation levels commonly in the 50% to 97% range. For practical synthesis, the strongest evidence supports solvent-based cement processing, optional co-olefin use, and ruthenium/osmium catalyst systems; catalyst loading and reaction time must be balanced to avoid gelation while achieving low residual unsaturation. For downstream performance, microgel-containing peroxide-cured formulations improve high-temperature mechanical properties, and functionalized HNBR derivatives can serve as fluorine-free battery binders. Overall, the recommended synthesis direction is controlled metathesis degradation of NBR to lower molecular weight, followed by selective hydrogenation under homogeneous catalytic conditions, with careful control of catalyst type, co-olefin, and hydrogenation extent to preserve processability and mechanical integrity.

5. REFERENCES

- US8124234B2 | Polyurethane compositions and articles prepared therefrom, and methods for ... | Not disclosed | US | 2012 | <https://patents.google.com/patent/US8124234B2/en>
- EP2125918B1 | Functionalized olefin polymers, compositions and articles prepared therefrom, ... | Not disclosed | EP | 2012 | <https://patents.google.com/patent/EP2125918B1/en>
- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- EP2057205B1 | Production of meta-block copolymers by polymer segment interchange | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2057205B1/en>
- US9321910B2 | Microgel-containing vulcanizable composition based on hydrogenated nitrile ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9321910B2/en>
- US20130345366A1 | Partially hydrogenated nitrile rubbers | Not disclosed | US | 2013 | <https://patents.google.com/patent/US20130345366A1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- EP3900086B1 | Electrode composition for a cathode of a cell of a lithium-ion battery, a ... | Not disclosed | EP | 2023 | <https://patents.google.com/patent/EP3900086B1/en>